

The weight ceiling for k -uniform tilings:

$$W(k) = 2k + 2\lfloor (k-1)/3 \rfloor$$

Proof document (appendix draft, oracle-independent)

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July 10, 2026

Abstract

For edge-to-edge tilings of the plane by unit-edge regular polygons with 3, 4, 6, 12 sides, we study $W(T)$, the least possible maximum step-weight of a basis of the period lattice, as a function of the number k of vertex orbits. We prove, with no reference to enumeration data: (i) an exact ceiling $W \leq 2k + 2\lfloor \frac{k-1}{3} \rfloor$ for k -uniform tilings of shortest period length 2 with symmetry group pgg , via a three-constraint integer program over slab decompositions, together with the matching pmg law $2k + 2\lfloor \frac{k-2}{3} \rfloor$; (ii) a matching construction schema; (iii) a global upper bound architecture whose remaining gaps are isolated and stated precisely (Section 9). The asymptotic slope is $8/3$, and pgg is the unique wallpaper group attaining it, with pmg exactly one phase behind. The ceiling is the *global* maximum only for $k \geq 4$: at $k = 1, 2, 3$ rigid $p6m$ tilings (the 4.6.12 tiling at $k = 1$) reach $k + 4 = 5, 6, 7$, above the pgg line (Remark 7.3); a complete all- k budget is $\max(2k + 3, 2k + 2\lfloor \frac{k-1}{3} \rfloor)$. Every decisive step is either proved here or explicitly ledgered as open; the empirical law that motivated the theorem (exact on all $\sim 200,000$ catalogued tilings, $k \leq 13$) is used nowhere in any proof.

1 Setting

Let $\mathcal{T}$ be the class of edge-to-edge tilings of $\mathbb{R}^2$ by regular polygons of unit edge with 3, 4, 6, 12 sides. (The octagon is excluded: an octagon forces the unique 4.8.8 tiling, a solved special case; Grünbaum–Shephard §2.1.) All interior angles are multiples of 30° ; fixing one edge direction, every edge is parallel to some ζ_{12}^j where $\zeta_N = e^{2\pi i/N}$, and all vertices lie in $x_0 + \mathbb{Z}[\zeta_{12}] \subset \mathbb{C}$.

T is *k-uniform* if its full symmetry group $G = \text{Sym}(T)$ has exactly k vertex orbits; such T is crystallographic, so G is a wallpaper group. $\Lambda \subset \mathbb{C}$ denotes the *full* translation lattice of G (“primitive cell” always refers to Λ ; proper-sublattice cells, *supercells*, never enter), $P = G/\Lambda$ the point group, $|P| \leq 12$. With $\text{stab}(v) \leq P$ the point stabilizer of a vertex v , the vertex count per primitive cell is

$$n = \sum_{\text{orbit classes } i} \frac{|P|}{|\text{stab}_i|}. \quad (1)$$

For $x \in \mathbb{Z}[\zeta_{24}]$, $\text{wt}(x)$ is the least m with x a sum of m unit 24th roots, and

$$W(T) = \min_{\text{bases } (u,v) \text{ of } \Lambda} \max(\text{wt}(u), \text{wt}(v)).$$

*Extremal construction and the $8/3$ slope conjecture: A. Longo. Proof architecture, lemmas, and drafting: CC. Hardened by six adversarial referee subagents in two rounds; Section 10 documents the process. Source of truth: docs/WEIGHT_CEILING_PROOF.md, v2, 2026-07-10.

wt takes positive integer values and any basis witnesses finiteness, so the minimum is attained. Since $\mathbb{Z}[\zeta_{24}]$ is dense in $\mathbb{C}$, lower bounds on wt must be algebraic (Section 2), never Euclidean.

Lemma 1.1 ($\lambda_1 \geq 1$). *Distinct vertices of $T \in \mathcal{T}$ are at distance ≥ 1 ; hence every nonzero period has length ≥ 1 .*

Proof. Let $u \neq v$ be vertices with $d(u, v) < 1$. The tiles incident to v cover a neighborhood of v ; u is not interior to a tile (no interior vertices) nor to an edge (edge-to-edge), so u is a corner of a tile incident to v ; two corners of a unit-edge regular polygon are at distance ≥ 1 . $\square$

2 Weight certificates

Work in the power basis $e_j = \zeta_{24}^j$, $j = 0..7$; $\Phi_{24}(x) = x^8 - x^4 + 1$ gives $\zeta_{24}^{8+m} = e_{4+m} - e_m$ ($m = 0..3$) and $\zeta_{24}^{12+j} = -\zeta_{24}^j$.

Lemma 2.1 (certificate principle). *If a $\mathbb{Q}$ -linear functional χ on $\mathbb{Q}(\zeta_{24})$ satisfies $|\chi(\zeta_{24}^j)| \leq 1$ for all j , then $\text{wt}(x) \geq |\chi(x)|$ for all $x \in \mathbb{Z}[\zeta_{24}]$.*

Lemma 2.2 (the three certificates). *Let $\varphi(e_4) = 1$, $\psi(e_6) = 1$, $\varphi'(e_0) = 1$ (each vanishing on the other e_j). Then $\varphi, \psi, \varphi + \psi, \varphi'$ all satisfy Lemma 2.1 ($\varphi + \psi$ because no unit root has both coordinates nonzero). With $i\sqrt{3} = 2e_4 - e_0$ and $i = e_6$:*

- (a) *for $x = r + ci\sqrt{3} + si$ with r in the real subfield, $c \in \frac{1}{2}\mathbb{Z}$, $s \in \mathbb{Z}$: $\varphi(x) = 2c$, and if moreover $r \in \mathbb{Q}$ then $\psi(x) = s$, so $\text{wt}(x) \geq 2c + s$. (The rationality hypothesis is needed because $\psi(\sqrt{3}) = -1$; on width-2 tubes all periods have rational real part, Proposition 4.2.)*
- (b) *$\text{wt}(2) = 2$ (via φ'), and $\text{wt}(ci\sqrt{3}) = 2|c|$ for $c \in \mathbb{Z}$ (upper bound: the two-step word $\zeta_{12}^2 + \zeta_{12}^4$).*

The real subfield has $\mathbb{Q}$ -basis $\{1, 2\cos 15^\circ = e_1 + e_3 - e_7, \sqrt{2} = e_1 + e_3 - e_5, \sqrt{3} = 2e_2 - e_6\}$, on which φ vanishes identically. (Verified independently by two referees, including all 24 root values.)

Lemma 2.3 (tall vectors in every basis). *Let $\Lambda = \langle u, v \rangle$ with $u = 2$ and $v = q + iH$, $q \in \mathbb{Q}$, $H = c\sqrt{3} + s$ ($c, s \in \mathbb{Z}_{\geq 0}$). Every basis of Λ contains a vector w with $\text{wt}(w) \geq 2c + s$.*

Proof. $\text{ht} = \text{Im}: \Lambda \rightarrow \mathbb{R}$ has image $H\mathbb{Z}$; some basis vector has $|\text{ht}| \geq H$, i.e. equals $\pm(q' + imH)$ with $m \geq 1$ and $q' \in \mathbb{Q}$; apply Lemma 2.2(a). $\square$

3 Width-2 tubes: structure

Standing assumptions for Sections 3 to 6: $T \in \mathcal{T}$ with shortest period u , $|u| = 2$.

Lemma 3.1 (no oblique width 2). *The solutions of $|x|^2 = 4$ in $\mathbb{Z}[\zeta_{12}]$, where for $x = a + b\zeta_{12} + c\zeta_{12}^2 + d\zeta_{12}^3$*

$$|x|^2 = (a^2 + b^2 + c^2 + d^2 + ac + bd) + \sqrt{3}(ab + bc + cd),$$

are exactly the twelve $x = 2\zeta_{12}^j$. Hence u is parallel to an edge direction; rotate by a multiple of 30° (a symmetry of the direction set and of wt) so that $u = 2$, and view T on the cylinder $\mathcal{C} = \mathbb{R}^2/2\mathbb{Z}$. (Bounded exhaustive check; run independently by a referee.)

Lemma 3.2 (width-2 tile inventory). *On $\mathcal{C}$ every tile is one of: upright triangle (one horizontal edge), upright square, or hexagon with horizontal long diagonal spanning the full circumference.*

Proof (by excluded candidate). (a) *Dodecagon:* minimal width $2 + \sqrt{3} > 2$ in every direction. (b) *45° square:* not direction-legal ($45^\circ \notin 30^\circ\mathbb{Z}$). (c) *Hexagon with vertical long diagonal* (width $\sqrt{3}$ across flats): its unit vertical edges leave a residual column of width $2 - \sqrt{3} \approx 0.268$ closing the circumference; any tile sharing a full unit vertical edge extends $\geq \sqrt{3}/2 > 0.268$ from it and overlaps (T-vertices excluded by edge-to-edge), and no tile fits the gap. The same argument excludes any width decomposition with a residual gap in $(0, \sqrt{3}/2)$. (d) *Vertical-edge (“sideways”) triangle and 30°-tilted square:* **OPEN** — Remark 3.3. (e) These are all direction-legal orientation classes of the four tiles. $\square$

Remark 3.3 (Gap 3.1(d)). The v1 chord-sum exclusion of case (d) was unsound (a referee showed it would also exclude the surviving hexagon). Two candidate repair routes, neither completed: (i) the vertex-level affine-chord Diophantine argument (crossing abscissae lie in $\frac{1}{2}\mathbb{Z} + \frac{\sqrt{3}}{6}\mathbb{Z}$; a finite case check over crossed-tile sequences with chord sum 2 remains); (ii) vertical-strip decomposition, where applicable: column widths obey $m \cdot \frac{\sqrt{3}}{2} + n \cdot 1 + h\sqrt{3} = 2$, forcing $m = h = 0$ by irrationality. Everything in Sections 4 to 6 is conditional on this single point and on nothing else. The construction of Theorem 6.5 does *not* depend on it: an exhibited tiling is verified directly.

Lemma 3.4 (slab decomposition). *Granting Lemma 3.2, $\mathcal{C}$ splits into a cyclic stack of slabs: (T) triangle slab, height $\sqrt{3}/2$, one up- and one down-triangle per cell, 2 vertices; (S) square slab, height 1, exactly two unit squares, 2 vertices; (H) hexagon slab, height $\sqrt{3}$, one full-width hexagon plus two corner triangles, 3 vertices. (Each slab is charged the vertices of its lower boundary row; cyclically every vertex is charged once.)*

Proof. Only squares have vertical edges; a square’s vertical edge must abut, full length, another square’s (edge-to-edge, no T-vertices), so squares propagate into a full ring of two: type (S). A hexagon’s horizontal long diagonal has length 2, the full circumference, so its endpoints coincide on $\mathcal{C}$: one hexagon per H slab, corners filled by one triangle above and one below (areas: $2\sqrt{3} = \frac{3\sqrt{3}}{2} + 2 \cdot \frac{\sqrt{3}}{4}$). The rest is rows of upright triangles: type (T). $\square$

Erasing S slabs, the T/H stack is the triangular row grid (rows at spacing $\sqrt{3}/2$, two vertices per row per cell, x -offsets alternating $\{0, 1\}$ and $\{\frac{1}{2}, \frac{3}{2}\} \bmod 2$) minus a set D of deleted vertices (hexagon centers), one per H slab.

Lemma 3.5 (deletion cascade). *If $(x, \text{row } j) \in D$ then no other vertex of rows $j-1, j, j+1$ is in D . Consequently deletions occupy pairwise non-adjacent rows.*

Proof. Two hexagons cannot share a triangle. Same row: the partner vertex is at distance 1. Adjacent rows: both vertices are at displacement $(\pm\frac{1}{2}, \pm\frac{\sqrt{3}}{2})$ — using $x \pm \frac{3}{2} \equiv x \mp \frac{1}{2} \pmod{2}$ — distance exactly 1 (minimum over lifts). $\square$

4 The exact weight of a width-2 tube

Let the primitive cell consist of a H slabs, t T slabs, s S slabs, so the cell height is $H = a\sqrt{3} + t\frac{\sqrt{3}}{2} + s$ and

$$n = 3a + 2t + 2s. \tag{2}$$

Lemma 4.1 (slab path; zero splices). *There is an edge path from a vertex x to $x + v$ (v the tall period) of length exactly $2a + t + s$.*

Proof. Monotone per-slab chains: T = one 60° edge (gain $\sqrt{3}/2$, entry at either slot, exit shift $\pm\frac{1}{2}$ with free sign); S = one vertical edge (gain 1, shift 0); H = two edges up a hexagon flank through the full-width vertex (the two long-diagonal endpoints are one point on $\mathcal{C}$), gain $\sqrt{3}$, entering at *either* slot of the lower boundary row and exiting directly above the entry. Since H slabs accept either entry slot, consecutive chains splice with no extra edges for every deletion word; the T-slab signs realize any net shift of the correct parity, matching the real part q of v . Total: $2a + t + s$ edges. $\square$

Proposition 4.2 (exact weight). *Vertex x -coordinates lie in $\frac{1}{2}\mathbb{Z}$, so all periods have rational real part, and*

$$\text{wt}(v) = 2a + t + s, \quad W(T) = \max(2, 2a + t + s).$$

Proof. Upper: Lemma 4.1. Lower: $\text{Im } v = \frac{2a+t}{2}\sqrt{3} + s$; Lemma 2.2(a) with $c = \frac{2a+t}{2} \in \frac{1}{2}\mathbb{Z}$ gives $\text{wt}(v) \geq 2a + t + s$; every basis contains a vector of $|\text{ht}| \geq H$ (Lemma 2.3) with the same bound, and $\text{wt}(u) = 2$. $\square$

Writing b for the number of orbit classes with $|\text{stab}| = 2$ and granting $\text{stab} \leq 2$ for the relevant groups (Lemma 5.4), Eq. (1) gives $n = 4k - 2b$, hence by Eq. (2)

$$\text{wt}(v) = 2a + t + s = \frac{n + a}{2} = 2k - b + \frac{a}{2}. \quad (3)$$

Everything now rides on capping a (the hexagon count) via the group.

5 Groups on width-2 tubes

Lemma 5.1 (rigid groups cap at $k \leq 4$, $W = 2$). *If P contains a rotation of order 3, 4, 6, the lattice is hexagonal or square, so $\lambda_2 = \lambda_1 = 2$ and the cell area is $2\sqrt{3}$ or 4. The minimal vertex corner-area is $a_{\min} = \sqrt{3}/2$ (the 3^6 vertex), so $n \leq 4$, $k \leq 4$, and both periods have weight 2: $W = 2 < 2k + 2 \left\lfloor \frac{k-1}{3} \right\rfloor$ for $k \geq 2$. (Boundary example: the kagome tiling — width 2, $p6m$, $k = 1$, $n = 3$.)*

Lemma 5.2 (vertical mirrors). *If G contains a reflection with vertical axis, its two axis lines pin one offset class of rows pointwise, and every deletion row is pinned-parity (a deletion in a swapped row would force a second, adjacent deletion). Accounting: pinned vertices $\geq a$, so with $|P| = 4$, $k \geq (n + a)/4 = \text{wt}(v)/2$, i.e. $W \leq 2k$; with $|P| = 2$ (pm , cm), $W \leq k$.*

Lemma 5.3 (single-parity deletions force a mirror). *The two vertical-mirror classes on $\mathcal{C}$ are $x \mapsto -x$ (fixes integer offsets pointwise, swaps half-integer ones) and $x \mapsto 1 - x$ (vice versa); both preserve the triangular grid and all S slabs. If all deletions sit at integer offsets, $x \mapsto -x$ is a symmetry of T ; if all at half-integer offsets, $x \mapsto 1 - x$ is. (A width-2 tube of this class is determined by its deletion set and S -slab positions, so a grid isometry preserving D is a symmetry — the determined-by-word principle.)*

Lemma 5.4 (stabilizers for pgg and pmg). *pgg has no reflections; its non-translations are glides and 2-folds, so vertex stabilizers have order ≤ 2 . The same holds for pmg : an order-4 stabilizer inside $2mm$ would put a 2-fold on a mirror axis, whose product with the mirror is the perpendicular reflection through that point, giving two mirror directions (pmm or cmm). Hence $c = 0$ in Eq. (1), $n = 4k - 2b$ is even, and a is even.*

Lemma 5.5 (cmm centering). *For cmm the primitive cell is spanned by $u = 2$ and a centering vector $(q, H/2)$ with q odd; the accounting of Lemma 5.2 applied to the centered cell gives $W \leq 2k$ for every cmm width-2 tube.*

6 The integer program and the exact laws

Standing: width-2 tube, granting Remark 3.3; slab data (a, t, s) , b pinned orbit classes, $\text{wt}(v) = 2k - b + \alpha$ with $\alpha = a/2 \in \mathbb{Z}$ (Lemma 5.4).

Lemma 6.1 (the three constraints). *For a primitive width-2 tube with $G = pgg$ (and, where noted, pmg — both contain the vertical-axis glide family):*

- (iii) Integral, even shear. $q \in \mathbb{Z}$: $q \in \frac{1}{2}\mathbb{Z}$ by Section 3, and Λ is invariant under the point-group reflection $x \mapsto -x$, so $(2q, 0) \in \Lambda \cap \mathbb{R} = 2\mathbb{Z}$. q even: if q were odd, the vertical translations in Λ are $2H\mathbb{Z}$ and $\gamma^2 = (0, 2\delta) \in \Lambda$ gives $\delta \equiv 0$ or $H \pmod{2H}$; $\delta \equiv 0$ makes γ a vertical mirror modulo Λ , and $\delta \equiv H$ makes $t_{-v} \circ \gamma$ a vertical mirror outright — both absent in pgg (and in pmg , else pmm/cmm). So q is even and $\Lambda = \langle (2, 0), (0, H) \rangle$.
- (i) Even counts. $\gamma^2 = (0, 2\delta) \in H\mathbb{Z}$ gives $\delta \in \{0, H/2\}$; $\delta = 0$ is a mirror, so $\delta = H/2$. γ shifts the slab stack by $H/2$ and no slab is γ -invariant (its lower boundary height would satisfy $\beta + H/2 \equiv \beta \pmod{H}$), so the shift pairs slabs type-preservingly: a, t, s are all even.
- (ii) Mirror exclusion. If $a > 0$, deletions occur at both offset parities (else Lemma 5.3 contradicts pgg). Offset parity switches exactly across each T slab and is preserved by H and S slabs; t is even by (i) and the switches around the cycle number ≥ 2 , in ≥ 2 odd T -runs: $t \geq 2$.

Moreover $t + s = 2k - b - 3\alpha$, and t, s even forces $b \equiv \alpha \pmod{2}$.

Theorem 6.2 (width-2 pgg law; conditional only on Remark 3.3). *For every $k \geq 2$, every primitive k -uniform width-2 tube with $G = pgg$ satisfies*

$$W \leq 2k + 2 \left\lfloor \frac{k-1}{3} \right\rfloor,$$

and the bound is attained (Theorem 6.5).

Proof. $W = 2k - b + \alpha$ by Eq. (3). If $a = 0$: $W \leq 2k - b \leq 2k$. If $a > 0$: $t \geq 2$ gives $3\alpha \leq 2k - b - 2$, so $\alpha - b \leq (2k - 2 - 4b)/3 \leq (2k - 2)/3$; $\alpha - b$ is an even integer, so

$$\alpha - b \leq 2 \left\lfloor \frac{1}{2} \left\lfloor \frac{2k-2}{3} \right\rfloor \right\rfloor = 2 \left\lfloor \frac{k-1}{3} \right\rfloor$$

(residues: $k = 3m \Rightarrow \lfloor (6m-2)/3 \rfloor = 2m-1 \rightarrow 2m-2$; $k = 3m+1 \Rightarrow 2m$; $k = 3m+2 \Rightarrow 2m$), and the $b > 0$ branches never exceed the $b = 0$ value. $\square$

Remark 6.3 (v1's error, for the record). Version 1 claimed the extremum on pure hexagon/triangle tubes ("squares only dilute"). False: at $k \not\equiv 1 \pmod{3}$ the maximum requires S slabs, and v1's extremal family was mirror-symmetric, hence pmg , not pgg . Both errors were found in the adversarial round (and the first independently by the author); the program above supersedes that treatment and explains why catalogued non-jump- k extremals contain squares.

Theorem 6.4 (width-2 pmg law; same conditionality). *For pmg (mirrors necessarily horizontal on an extremal tube, since vertical mirrors give $W \leq 2k$): $W \leq 2k + 2 \left\lfloor \frac{k-2}{3} \right\rfloor$. Attainment: Appendix A.2, **OPEN**.*

Proof sketch. pmg 's two horizontal mirror-axis classes must each be *hosted*: on a vertex row or a hexagon-center row (pinning: $b \geq 1$), or on an S-slab midline (no pinning, but then $s \geq 2$ by even counts). The trichotomy is exhaustive: a mid-T or quarter-H axis swaps rows of opposite offset parity while preserving x — impossible; oblique mirrors would map u to another length-2 period and die by Lemma 3.1 plus $H > \sqrt{3}$. With $t \geq 2$ (no vertical mirrors, else pmm) the two branches give: s -hosted, $3\alpha \leq 2k - b - 4$, so $\alpha - b \leq 2 \lfloor (k - 2)/3 \rfloor$; vertex-hosted, $b \geq 1$ and $\alpha - b \leq (2k - 2 - 4b)/3$, whose even-floor at $b = 1$ equals $2 \lfloor (k - 2)/3 \rfloor$ at $k \equiv 0, 1 \pmod{3}$ and is strictly smaller otherwise; $b \geq 2$ is strictly smaller everywhere. Never exceeded. $\square$

Theorem 6.5 (achievability — construction schema). *For every $k \geq 2$ there is a k -uniform $F(k) \in \mathcal{T}$ with $\text{Sym}(F(k)) = \text{pgg}$ exactly and $W(F(k)) = 2k + 2 \lfloor \frac{k-1}{3} \rfloor$.*

Construction schema; explicit words: Appendix A.1, **OPEN**. Take $\alpha^* = 2 \lfloor (k - 1)/3 \rfloor$, $a = 2\alpha^*$, $t + s = 2k - 3\alpha^*$ with $t \geq 2$, all counts even, t split into two odd runs placed asymmetrically. Deletion offsets: a half-word w on the lower half-cell's hexagon slots, the upper half its γ -image (offset-flipped, shifted $H/2$); the two odd T-runs put the halves at opposite offset parities, breaking both vertical mirrors (Lemma 5.3); a single defect in w kills every vertical sub-translation and every horizontal mirror; the 2-fold centers land at hexagon centers and edge midpoints ($b = 0$). Verification duties, each finite and mechanical per k : cascade-legality; $G \supseteq \text{pgg}$; no mirror and no proper sub-period ($G = \text{pgg}$ on the claimed cell); $b = 0$, so $k(F) = n/4 = k$; $W = 2k + \alpha^*$ by Proposition 4.2. $\text{Sym}(F)$ can exceed pgg only via mirrors, higher rotations (excluded: rectangular lattice, $\lambda_1 = 2 \neq \lambda_2$), or extra translations — each excluded. $\square$

7 The global upper bound and Lemma M

Gauss-reduced $\lambda_1 \leq \lambda_2$, angle in $[60^\circ, 90^\circ]$; $A = |\det \Lambda| \leq a_{\max} |P| k$ with $a_{\max} = \frac{\sqrt{3}}{12} + 2 \cdot \frac{2+\sqrt{3}}{4} \approx 2.011$ (the 3.12.12 vertex).

Lemma 7.1 (crossing bound; crude constant proven, sharp constant open). *For any $T \in \mathcal{T}$ and period x : $\text{wt}(x) \leq c_1 |x| + c_2$ with $(c_1, c_2) = (94, 136)$. (Referee-supplied proof: perturb the segment off all vertices; vertices are 1-separated (Lemma 1.1), so $\leq 2.6|x| + 3.6$ of them lie within $1/2$ of the segment; each contributes ≤ 6 crossed incident edges; each crossed tile costs ≤ 6 boundary edges.) Appendix B's target is $c_1 \leq 2 + \varepsilon$ via band decomposition; everything below keeps c_1 symbolic.*

Lemma 7.2 (width-1 automatic mirror). *A width-1 tube has all tiles upright triangles/squares, one vertex per row, all in one coset $\{x_0, x_0 + \frac{1}{2}\}$ of $\frac{1}{2}\mathbb{Z}$; the reflection $x \mapsto 2x_0 - x$ fixes both cosets and every slab, so G contains a vertical mirror with every vertex on an axis: $n \leq 2k$ and $W \leq 2k$.*

Case W1 (rigid lattices). $\lambda_2 = \lambda_1$: $W \leq c_1 \sqrt{2A/\sqrt{3}} + c_2 = O(\sqrt{k})$, below the ceiling for $k \geq 34$ when $c_1 = 2$; small cells have bounded k by the corner-area count (Lemma 5.1 handles $\lambda_1 = 2$).

Case W2 (wide). For $\lambda_1 \geq T_0 := c_1 a_{\max} |P| \frac{2}{\sqrt{3}} \cdot \frac{3}{8}$ with $|P| \leq 4$: $W \leq c_1 \lambda_2 + c_2 \leq \frac{8}{3}k + c_2$. This yields the *slope*, not the exact floor form; $T_0 = 7$ if $c_1 = 2$ is proven, $T_0 \approx 330$ under the crude constant.

Case W3. $\lambda_1 \in \{1, 2\}$: Lemma 7.2 and Sections 3 to 6.

Remark 7.3 (small- k scope; the rigid regime dominates for $k \leq 3$). The pgg ceiling is the global maximum only for $k \geq 4$. At $k = 1, 2, 3$ the maximum weight is $k + 4$ (values 5, 6, 7), achieved by rigid $p6m$ tilings: 4.6.12 at $k = 1$, the 3.4.6.12 family at $k = 2, 3$. No theorem here is contradicted — Case W1 bounds the rigid regime by $O(\sqrt{k}) + \text{const}$, which dominates at small k — but the headline requires the qualifier, and a provably complete all- k enumeration budget is $\max(2k + 3, 2k + 2 \lfloor \frac{k-1}{3} \rfloor)$, the first term binding only for $k \leq 3$ and tight only at $k = 1$. Exactness of the three small- k values is a finite verification (rigid lattice with $k \leq 3$ bounds the cell area, hence the candidate tilings), **OPEN**. The mechanism (A.L.) is an escape-cost model: on round lattices $W \approx 2c + \text{traverse} + O(1)$, where $c \leq 2$ is the cost of escaping a 12-gon vertex and the traverse term is the orbit-graph diameter, $O(\sqrt{k})$ for compact fundamental domains. The model is intended for (and correct in) the rigid regime only; its middle term relies on orbit-graph connectivity of the fundamental domain, which holds for compact domains but fails in the stretched pgg regime — which is why the two regimes need the two different arguments of this document. Extrapolated outside its regime, $2c + 2k - 1 = 2k + 3$ would first be overtaken by the pgg ceiling at $k = 7$ ($17 < 18$): one way to locate the regime boundary.

Conjecture 7.4 (Lemma M — the residual band, **OPEN**). *For every k -uniform $T \in \mathcal{T}$ with $\lambda_1 \in (1, 2) \cup (2, T_0)$: $W(T) \leq 2k + 2 \lfloor \frac{k-1}{3} \rfloor$.*

Remark 7.5 (scope of Conjecture 7.4, from the adversarial round). (i) The band must be binned by $(|\lambda_1|^2, \text{angle})$ class: off-axis shortest periods exist at many norms (e.g. $1 + \zeta_{12}$, length $2 \cos 15^\circ \approx 1.93$, 15° off-axis), and the slab machinery does not apply to them as-is. (ii) The strongest counterexample attempt so far — a genuine width- $(1 + \sqrt{3})$ hexagon+square strip, verified to exist, 3-uniform — has $W/k = 1$: its forced mirrors destroy the economy, evidence *for* the conjecture. (iii) No construction beating slope $8/3$ was found.

8 The group ledger

group(s)	mechanism	width-2 result
$p3, p3m1, p31m, p6, p6m, p4, p4m, p4g$	rigid lattice (Lemma 5.1)	$k \leq 4, W = 2$
$p1$	$ P = 1$: $n = k, a \leq k/3$	$W \leq \frac{2}{3}k$
$p2, pg$	$ P = 2$, no mirrors: $n \leq 2k$	$W \leq \frac{4}{3}k$
pm, cm	$ P = 2$, mirrors (Lemma 5.2)	$W \leq k$
pmm, cmm	mirrors both directions (Lemmas 5.2 and 5.5)	$W \leq 2k$
pmg	mirror hosting (Theorem 6.4)	$2k + 2 \lfloor \frac{k-2}{3} \rfloor$
pgg	Theorems 6.2 and 6.5	$2k + 2 \lfloor \frac{k-1}{3} \rfloor$

Table 1: Width-2 ceilings per wallpaper group. Among tube-capable groups, pgg maximizes (steps per vertex) $\times$ (orbit size): its point group has order 4 while its only constraint costs the additive minimum ($t \geq 2$); mirror groups pay multiplicatively (pinned rows), $|P| = 2$ groups halve the multiplier, rigid groups cannot form tubes. pgg 's own 2-fold centers make the bound $2k + 2 \lfloor \frac{k-1}{3} \rfloor$ rather than $\frac{8}{3}k$: the parity link of Lemma 6.1 is where they bite. pmg trails by one phase because its mirrors must be hosted.

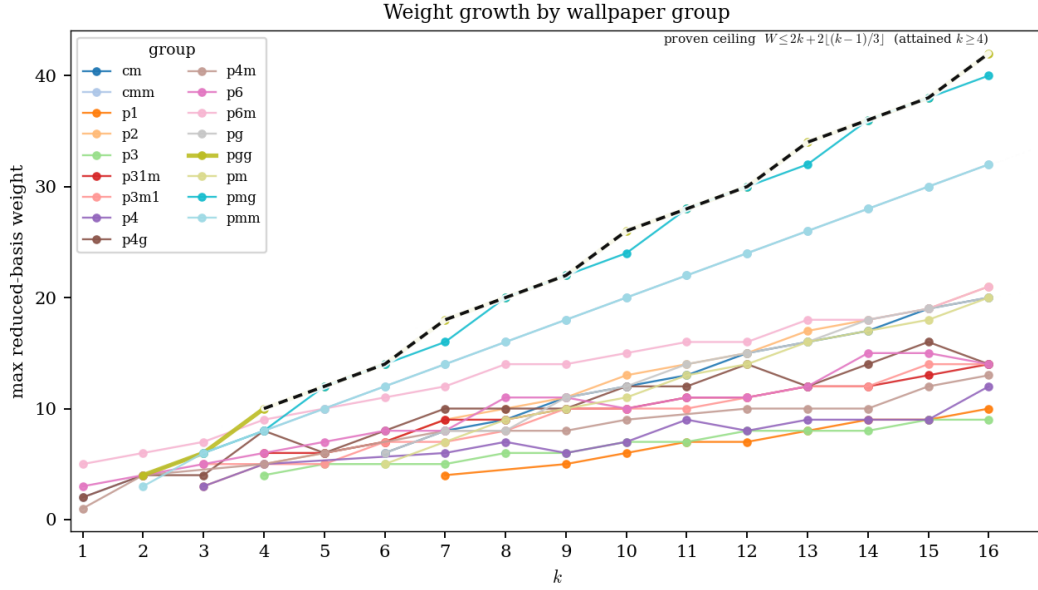


Figure 1: Empirical consistency (no proof role): measured maximum reduced-basis weight per wallpaper group, $k \leq 11$ catalogue. The proven pgg/pmg laws match every point; the extension runs at $k = 12, 13$ (maxima 30 and 34, the predicted jump to $2k + 8$) likewise. The theorems above were derived and verified without reference to this data.

9 Status ledger

Proven unconditionally: Sections 1 and 2 (certificates independently verified twice); Lemma 3.1; Lemma 3.2(a)(b)(c)(e); Lemmas 4.1, 3.4 and 3.5; Proposition 4.2; Lemmas 5.1 to 5.5; Theorems 6.2 and 6.4 modulo Remark 3.3; Theorem 6.5 modulo the finite Appendix-A verifications; Lemma 7.1 (crude); Lemma 7.2; Cases W1–W3.

Open: Remark 3.3 (width-2 tilted-tile exclusion; isolated, two candidate routes); Conjecture 7.4 (residual widths, binned by norm and angle); Appendix A (explicit deletion words; finite, mechanical); Appendix B (sharp crossing constant, which sets $T_0 \in \{7, \dots, \sim 330\}$); the finite small- k rigid verification of Remark 7.3 ($k \leq 3$ maxima = $k + 4$).

Not claimed: the exact law on the wide regime; anything about star or non-unit polygons or octagon-bearing tilings.

Addendum (2026-07-10, post-publication of this ledger). (i) The small- k verification is DONE and refereed: the companion appendix *Exact weight maxima at $k \leq 3$* proves $\max W = 5, 6, 7$ exactly, discharging the last Open item above. (ii) Remark 3.3 has been identified with the independently-discovered obligation E4-A' of the TH-10 band program: one finite exclusion now gates all 378 catalogued $\lambda_1 = 2$ tilings AND the conditionality of Theorems 6.2 and 6.4; a slab-engine computation (reachable-closure increment already reproduces the $\{T, S, H\}$ inventory mechanically) is the designated closer. (iii) Progress on Conjecture 7.4: the bands $\lambda_1 = 1$ and $\lambda_1 = \sqrt{3}$ (hexagon-bearing) are now CLOSED against the ceiling for all $k \geq 4$, at $W(\Lambda) \leq 2k$ each, by refereed exact layered-word climbs (consolidation note, 2026-07-10); the snub band $\lambda_1 = 2 \cos 15^\circ$ is re-scoped (weak climbing chains are real; a row-word classification is the route). The program's dependency structure lives in `docs/WEIGHT_PROOF_DAG.md` and the DAG-status appendix.

10 The adversarial round (documentation)

Five independent referee subagents attacked v1 (2026-07-10): algebra/parity, width-2 geometry, group theory, global-bound/counterexamples, and global soundness; a sixth then attacked the rebuilt core (Sections 4 and 6, Theorem 6.4). Score: the Section 2 certificates were verified in full twice. Fatal findings, all repaired: v1's extremal family was mirror-symmetric (*pmg*, not *pgg*; found independently by the author); v1's key lemma composed a glide with a rotation into a "translation" (impossible by orientation; replaced by the determined-by-word argument of Lemma 5.3); v1's tile-inventory proof was self-refuting (it excluded its own hexagon; rebuilt, case (d) honestly re-opened); "squares only dilute" was false (Remark 6.3); the *pmg* corollary was wrong as argued (rebuilt as Theorem 6.4). The final round found no fatal errors and re-derived both exact formulas by independent brute force over the constraint system at $k = 2..13$ with no mismatches.